

SMART MUSEUM RESOURCE MANAGEMENT PLATFORM

SMRMP

Full System Functionality Document
Product Requirements Document (PRD)

Document Type	System Functionality & Role-Based PRD
Version	2.0
Status	Active Development Reference
Prepared	July 2026
Reference Site	Adwa Victory Memorial Museum, Ethiopia
Backend	Node.js + Express.js + PostgreSQL
Frontend	React.js + Tailwind CSS
AI Layer	OpenAI API

ARTIFACT INTELLIGENCE · CONSERVATION · DIGITAL VISITOR EXPERIENCE · HERITAGE ANALYTICS

1. Table of Contents

1. Table of Contents
2. Executive Summary
3. System Purpose & Scope
4. Technology Architecture
5. User Roles & Access Control (RBAC)
6. Module Functionality (Full Catalog)
7. Key Business Workflows
8. Artificial Intelligence Capabilities
9. Payments, QR & Integrations
10. REST API Reference
11. Security & Data Governance
12. Implementation Roadmap
13. Success Criteria & KPIs

2. Executive Summary

The **Smart Museum Resource Management Platform (SMRMP)** is an AI-assisted operations platform that digitizes museum artifact catalogs, conservation work, exhibition planning, visitor ticketing, and executive analytics. It is designed for institutions that still rely on paper ledgers and disconnected spreadsheets, with **Adwa Victory Memorial Museum (Ethiopia)** as the reference deployment.

SMRMP acts as a **decision-support and record-of-truth system**. AI drafts descriptions, flags risk, and answers operational questions — but every AI output is intended for human curator or manager review before it becomes part of the official record.

Core value delivered

- Complete digital identity for every artifact (metadata, images, QR code, history).
- Structured conservation logs tied to each artifact.
- Exhibition planning and artifact allocation.
- Digital ticketing with Telebirr / Chapa / cash (sandbox) and gate QR verification.
- Role-based staff access (admin, curator, conservation, maintenance, researcher, visitor).
- Live dashboards for collection, conservation, and visitor metrics.
- AI assistants for description drafting, smart search, reports, and Q&A;

3. System Purpose & Scope

3.1 Problem addressed

Problem Area	Challenge	Impact
Artifact Management	Manual registration; incomplete docs; weak location/custody history	High
Operations	Departmental silos; hand-compiled reports; lost maintenance requests	Medium
Visitor Experience	Cash-heavy ticketing; limited self-guided digital context	Medium
Decision-Making	No consolidated analytics; risks found only after damage or loss	High

3.2 Objectives

Objective	Success Looks Like
Digital Transformation	Paper workflows replaced by centralized, role-based digital records
Artifact Preservation	Every artifact has searchable documentation and auditable condition history
Operational Efficiency	Shared, trackable workflows for inventory, maintenance, and staff coordination
Visitor Engagement	QR exploration and digital ticketing without counter-only dependence
Intelligent Decisions	Live dashboard + AI-drafted reports as starting points for leadership review

3.3 In scope (implemented MVP)

- Authentication (Supabase Auth + app users table), password flows, forced staff password change
- Full RBAC with system & custom roles and permission matrix
- Artifact CRUD, Cloudinary images, QR identity, public QR lookup, audio narration UI
- Exhibition CRUD
- Conservation log CRUD
- Ticketing: types, purchase, verify, Telebirr/Chapa/cash sandbox
- Dashboard stats & charts
- AI: describe artifact, smart search, generate report, ask assistant
- Admin: users, roles, permissions

- Audit logging support

3.4 Out of scope / Phase 2–3

- Inventory / barcode stock management
- Staff & volunteer HR (attendance, leave, shifts)
- Facility maintenance work orders
- Full digital archive with version control
- Memberships, school/group bookings
- RFID hardware, IoT climate sensors, AR/VR tours
- Multi-museum national multi-tenant platform
- Remote Desktop Protocol (RDP) — **not part of SMRMP**; this system is a web museum platform, not remote desktop software

Note: If “RDP” was intended to mean this Product Requirements Document (PRD), this PDF is that deliverable — full system functionality in PRD form.

4. Technology Architecture

4.1 High-level layers

Visitors and staff use browsers (and mobile QR). The **React SPA** talks to a **Node.js / Express REST API**, which uses **PostgreSQL (Sequelize)**, **Supabase Auth**, **OpenAI**, **Cloudinary**, and **Telebirr/Chapa** sandboxes.

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS, React Router 6, Axios, TanStack Query, Zustand, Chart.js, html5-qrcode
Backend	Node.js, Express 4, Sequelize ORM, JWT, Multer, Helmet, Morgan, express-rate-limit, express-validator
Database	PostgreSQL (local or Supabase-hosted)
Auth	Supabase Auth + public.users with RBAC (roles / permissions)
Media	Cloudinary (artifact images)
AI	OpenAI API
Payments	Telebirr (sandbox), Chapa, cash (simulated)
Identity codes	QR via qrcode library (artifacts + tickets)

4.2 Repository layout

- **smrmp-frontend/** — Vite React SPA (landing, auth, artifacts, exhibitions, tickets, admin, visitor, dashboard)
- **smrmp-backend/** — Express API, migrations, seeders, tests, services
- **docs/** — RBAC notes and this functionality PDF
- **README.md** — Role-Based PRD (development reference)
- **SMRMP_Proposal.md** — Concept & solution proposal

5. User Roles & Access Control (RBAC)

Access is permission-based (`requirePermission('artifacts.read')`), not hardcoded role lists. Login and `/auth/me` return `permissions[]` for frontend `can()` checks.

5.1 System roles

Role	Purpose	Typical access
admin	Full system access	All permissions (protected perms cannot be stripped)

Role	Purpose	Typical access
curator	Catalog & exhibitions	Artifacts/exhibitions/conservation write, tickets, dashboard, AI
conservation	Condition & restoration	Artifacts read, conservation CRUD, tickets verify/list, dashboard
maintenance	Facilities / gate	Tickets list/verify + dashboard
researcher	Read-only research	artifacts.read only
visitor	Public account	No staff permissions; public signup

5.2 Account creation paths

Path	Who	Resulting role
POST /api/auth/register	Public visitor signup	Always visitor
POST /api/users	Admin (users.create)	Staff role + temp password; must_change_password=true

5.3 Permission modules

Permission codes follow module.action: users.*, roles.*, artifacts.*, exhibitions.*, conservation.*, tickets.*, dashboard.read, ai.describe / ai.search / ai.report / ai.ask.

Admin UI: /admin/users, /admin/roles, /admin/permissions.

6. Module Functionality (Full Catalog)

6.1 Authentication & session

- Register, login, logout, session profile (/me)
- Forgot password, change password, update password
- Forced password change for staff on first login
- Supabase-backed authentication integrated with app RBAC

6.2 User & role administration

- Create staff accounts and assign roles
- Activate / deactivate users
- Create custom roles and edit permission matrix
- System roles cannot be deleted; admin protected permissions retained

6.3 Artifact management (Module 1 — Priority)

- CRUD for artifacts: name, category, period, origin, material, description, location, condition
- Multi-image upload via Cloudinary
- Unique QR digital identity per artifact
- Public lookup by QR code (/artifact/:code) without staff login
- AI-drafted descriptions for curator review
- Duplicate detector UI, condition/location history UI, audio narration player
- Visitor flow: Scan QR → Artifact profile → History & provenance → Gallery → Audio

6.4 Exhibition management (Module 2 — Priority)

- Create, schedule, update, and archive exhibitions
- Gallery / space assignment
- Artifact allocation to exhibitions
- Historical exhibition archive and performance analysis (attendance/engagement — roadmap)

6.5 Conservation management (Module 3 — Priority)

- Conservation logs tied to artifacts and inspectors
- Inspection / condition / restoration history per artifact
- Damage reporting workflow
- Dashboard surfaces conservation alerts
- Environmental sensor feed planned for Phase 2 (manual entry initially)

6.6 Ticketing & visitor payments

- Public ticket type listing and purchase
- Staff CRUD for ticket types
- Digital QR tickets after purchase
- Gate verification by ticket code (tickets.verify)
- Payment methods: Telebirr, Chapa, cash (sandbox / simulated)

6.7 Executive dashboard (Module 9 — Priority)

- Stats: artifacts, exhibitions, conservation, visitors, revenue indicators
- Charts for trends and status snapshots
- Permission-gated: dashboard.read

6.8 AI assist layer

- Describe artifact — draft text from notes/images
- Smart search — natural-language catalog queries
- Generate report — operational summary drafts
- Ask assistant — grounded Q&A; over platform data drafts
- Rate-limited; outputs labeled for human review

6.9 Visitor public experience

- Marketing landing site
- Visitor registration
- Public artifact page via QR
- Ticket purchase without staff counter dependence

6.10 Audit logging

- AuditLog model and middleware support for sensitive changes
- Especially important for conservation and location updates

6.11 Phase 2–3 modules (proposed)

- Module 4 — Inventory (equipment, supplies, barcode stock, procurement)
- Module 5 — Staff & volunteer management
- Module 6 — Maintenance work orders & building systems
- Module 7 — Digital archive (documents, media, versioned policies)
- Module 8 — Memberships, group bookings, feedback
- IoT climate monitoring; RFID tracking; multi-tenant national network

7. Key Business Workflows

7.1 Catalog an artifact

Staff creates artifact → uploads images → system generates QR → optional AI description → curator reviews → record publishable; public can open via QR.

7.2 Visitor QR explore

Visitor scans physical QR → /artifact/:code → public profile with gallery, history, and narration.

7.3 Buy and verify a ticket

Visitor selects ticket type → pays via Telebirr/Chapa/cash (sandbox) → receives digital QR ticket → staff verifies at gate with tickets.verify.

7.4 Conservation inspection

Conservation staff logs inspection/condition against artifact → history accumulates → dashboard highlights items needing attention.

7.5 Plan an exhibition

Curator creates exhibition, sets schedule/space, allocates artifacts → team views shared schedule.

7.6 Admin onboarding staff

Admin creates user with role and temp password → staff logs in → forced password change → permissions drive UI and API access.

7.7 AI-assisted operations

Staff requests description/search/report/Q&A; → AI returns draft grounded in data → human reviews before official use.

8. Artificial Intelligence Capabilities

Capability	What it does	Human checkpoint
AI Artifact Assistant	Drafts descriptions, summaries, keywords, classification suggestions	Curator approves before publishing
AI Smart Search	NL search e.g. "Ethiopian artifacts from the Adwa period"	Ranked suggestions only — no autonomous edits
AI Report Generator	Drafts daily/monthly/executive summaries from live data	Manager reviews and signs off
Predictive Conservation	Flags elevated risk from condition history (Phase 2)	Conservation lead confirms schedule
Duplicate Detection	Flags likely duplicate catalog entries	Curator confirms merge or keep separate
AI Museum Assistant	Answers ops questions grounded in platform data	Staff verify before acting on answers

Constraint: AI quality depends on data completeness. Early outputs should be treated as first drafts and clearly labeled as AI-generated.

9. Payments, QR & Integrations

- **Payments:** Telebirr, Chapa, and cash simulation for tickets (donations/memberships planned).
- **QR:** Low-cost artifact identity and visitor digital tickets; printed labels near zero hardware cost.
- **RFID:** Phase 2/3 for higher-value inventory verification.

- **IoT:** Temperature, humidity, light, air quality → conservation alerts (future).
- **Media:** Cloudinary for artifact imagery.
- **Auth:** Supabase Auth.
- **AI providers:** OpenAI; regional options (e.g. Addis AI) and TTS for narration may be evaluated later.

10. REST API Reference

Base path: /api. Authenticated routes require a valid session/token; permission-guarded routes require the listed permission.

Area	Endpoints (summary)
Auth	POST /auth/register, /login, /logout; GET /auth/me; POST /auth/change-password, /forgot-password, /update-password
Artifacts	GET/POST /artifacts; GET/PUT/DELETE /artifacts/:id; public GET /artifacts/qr/:code
Exhibitions	GET/POST /exhibitions; GET/PUT/DELETE /exhibitions/:id
Conservation	GET/POST /conservation; GET/PUT/DELETE /conservation/:id
Dashboard	GET /dashboard/stats; GET /dashboard/charts
Tickets	CRUD /tickets/types; POST /tickets/purchase; GET /tickets; GET /tickets/:id; PATCH PUT DELETE /tickets/:id; GET /tickets/verify/:code
AI	POST /ai/describe-artifact, /ai/search, /ai/generate-report, /ai/ask
Users	GET/POST /users; PATCH /users/:id; PATCH /users/:id/status
Roles	GET /roles; GET /roles/permissions; POST/PATCH/DELETE /roles/:id; PUT /roles/:id/permissions
Health	GET /health

10.1 Primary frontend routes

/, /login, /register, /artifact/:code, /tickets*, /dashboard, /artifacts*, /exhibitions/*, /admin/*

11. Security & Data Governance

- Role-based access control aligned to museum stakeholder roles
- Secure authentication with password change policies for staff
- Audit logging for sensitive record changes
- Helmet, CORS, rate limiting, request validation on the API
- Regular backups with defined RPO/RTO (operational policy)
- Human approval trails for conservation sign-off and AI-assisted reporting
- Data residency and image/metadata ownership policy required before production cloud choice

12. Implementation Roadmap

Phase	Horizon	Focus
Phase 1 — MVP	Hackathon / Day 1	Auth, artifacts, AI assist, dashboard, QR visitor page, payment prototype
Phase 2 — Pilot	Months 1–6	Harden Modules 1–3; inventory/maintenance/staff; live Telebirr; real collection migration
Phase 3 — Scale	Months 6–18	Multi-tenant, mobile apps, RFID/IoT for funded sites, national heritage network

13. Success Criteria & KPIs

- % of collection digitized with complete required metadata and images
- Conservation follow-up completion rate (inspections logged on schedule)

- Staff adoption: active weekly users by role
- Visitor digital ticket share vs counter-only sales
- QR scan engagement on public artifact pages
- Time to produce monthly operational report (manual vs AI-assisted draft)
- Gate verification success rate and ticket fraud incidents

— End of Document —

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